

WEEK 4 STATUS REPORT: TRAFFIC HANDLING, BUFFER TUNING, & QOS STREAMING

1. Stream Optimization & Terminal Tuning

- 1 Evaluated the data transmission patterns of Cisco IOS streaming over secure network sockets.
- 2 Integrated the '**terminal length 0**' inline configuration command immediately following channel initialization. This temporarily deactivates terminal output pagination, successfully mitigating the automation-breaking '--More--' pause condition.
- 3 Established a mandatory 3-second application sleep state threshold following command dispatch. This provides a clean execution window for dense data strings to travel across the network before the application samples the stream.

2. Buffer Capacity Management

- 1 Upgraded the default socket read buffer size parameters to an explicit max-threshold window of 65,535 bytes (**channel.recv(65535)**).
- 2 This allocation prevents configuration truncation and buffer overruns, guaranteeing total data integrity for long configuration files.
- 3 Confirmed full end-to-end data integrity across R1, R2, and R3 backup log structures.